Storyline outline — zonality of soil-carbon turnover

Working document for comment (not the manuscript)

2026-06-26

This is the operational story bible: the narrative bones, the four pillars, the figures we plan to use (with comments), and the key numbers — all backed by repeatable scripts. Numbers are the production values (500-tree models, MAIN 6-layer biology) unless noted. Its companion is `results_reconciliation_2026-06-26.md` *(number map, figure dispositions, manuscript checklist). Read this; then we write the manuscript together.*

1 Thesis and contribution

One sentence. Beyond climate, soil-carbon turnover carries a coherent, attributable, and mappable spatial organisation — modest in magnitude but real — that Earth System Models (ESMs) over-represent and disagree on; we offer it both as an interpretive framework and as an observational benchmark.

That climate does not fully control soil carbon is not new (Schmidt et al. 2011 made the case a decade ago). *Our* contribution is not the *existence* of a beyond-climate signal but its *organisation*: we show the residual is spatially structured rather than noise, we attribute it to mechanistic process classes in an order-independent way, and we map where it acts. The framing is deliberately tool-first — a way of *reading* non-climate turnover structure — with the ESM contrast as the closing "so what", rather than a paper whose entire weight rests on a model critique.

A second, honest pillar of the contribution is restraint. The beyond-climate signal is real but small (∼7 percentage points of explained variance), most of the point-scale residual is unstructured noise, and the organised part only emerges at coarse (meso) scale. We treat this modesty as a feature: every claim is made on a spatially-honest footing and survives a battery of robustness checks.

2 Narrative spine (OCAR)

Opening. Soil-carbon turnover is routinely treated as climate-dr[illegible] with the remainder dismissed as noise or left unresolved. Yet on a *spatially-honest* basis [illegible]lock cross-validation that prevents nearby samples from leaking between training and test — climate explains only about *one third* of the predictable variance in turnover time.

Challenge. Is the other two thirds merely noise, or is it (i) spatially *organise*[illegible]) *attributable* to identifiable process classes, and (iii) *mappable*? And if it is organised, do the ESMs we rely on for projections represent that organisation correctly?

Action. We answer with four complementary analyses on one common dataset — an order[illegible] independent variance attribution (Shapley), a test of spatial organisation (Moran's I), a meso scale zonality map, and mechanistic response curves (ALE) — each wrapped in an honesty apparatu[illegible]patial block-CV, a noise-floor analysis, a provenance control, a biological-parsimony check, and a block-size sensitivity).

Resolution. The beyond-climate signal is[illegible]herent, ecologically-interpretable zonality: edaphic-led, with soil biology forming a distinct second[illegible]. It is modest in size but spatially organised and ecological in origin (not a sampling artifa[illegible] ESMs amplify it far too steeply toward the poles (∼3.6× on average) and disagree among themselves by an order of magnitude. We deliver the pattern as a framework for interpretation and a benchmark for models.

3 The four pillars

Each pillar answers a distinct question and has a dedicated figure. Together they make framing A concrete: *how much* (attribution), *where* (map), *how* (mechanism), and *so what* (ESM contrast).

3.1 Attribution — "how much?" (Shapley)

We fit nested Random Forests on a *single common row set* (so coalition R^2 are mutually comparable) and decompose the spatially cross-validated predictable variance with Shapley values, which average each domain's contribution over all orderings and therefore avoid the arbitrariness of nested increments. The split is Climate 35%, Edaphic 33%, Biological 22%, LandUse 10% — i.e. climate is only about a third, and non-climate domains carry two thirds. Strikingly, edaphic properties predict turnover *as well as climate does on their own* (single-domain R^2 tie at 0.31). The single domains sum to far more than the full model (0.95 vs 0.38), a large overlap of 0.57: the domains are *correlated, not redundant*. This is the conceptual keystone — it is why a per-pixel "which-process-wins-here" map fails (the channels are too correlated to argmax) while the aggregate attribution succeeds.

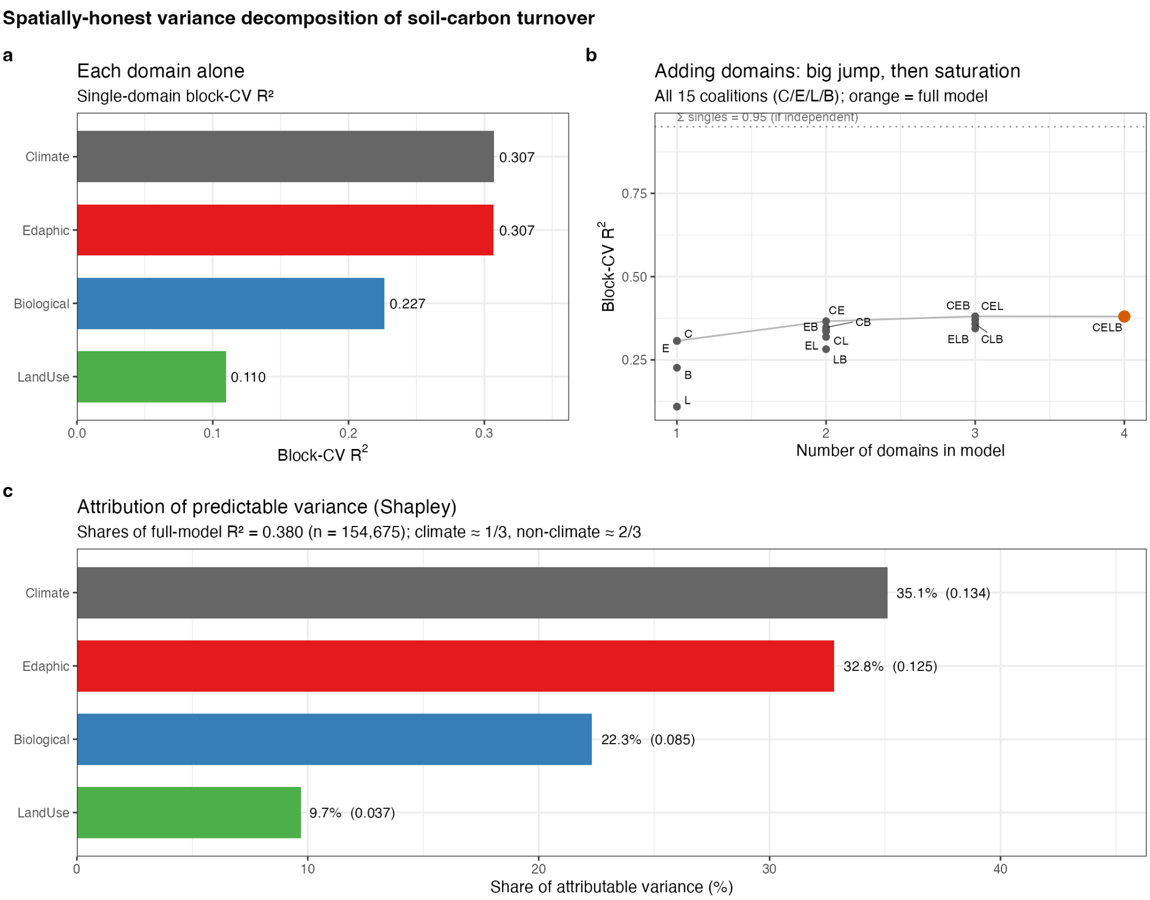

Figure 1: **Variance decomposition (Results).** Singles → build-up across coalitions → Shapley shares. The build-up panel makes the overlap visible (the full model sits far below the sum-of-singles line). `13c_results_figure.R`

3.2 Zonality map — “where?”

We map the log-ratio of nested model predictions — how much non-climate context bends predicted turnover relative to a simpler model — at meso (2°) scale, because the residual is two-thirds noise at the native pixel. The *symmetric* formulation (each domain taken relative to climate; no abiotic-before-biology ordering) is the main figure, consistent with the order-independent Shapley attribution. The abiotic panel is near-global over soil-bearing land and shows a coherent geography: non-climate context *lengthens* turnover in the Amazon and Congo basins (deep, weathered, low-fertility clays that stabilise carbon) and *shortens* it across drylands (Sahel margins, central Asia, Australia). This is the pattern that earns the word “zonality.”

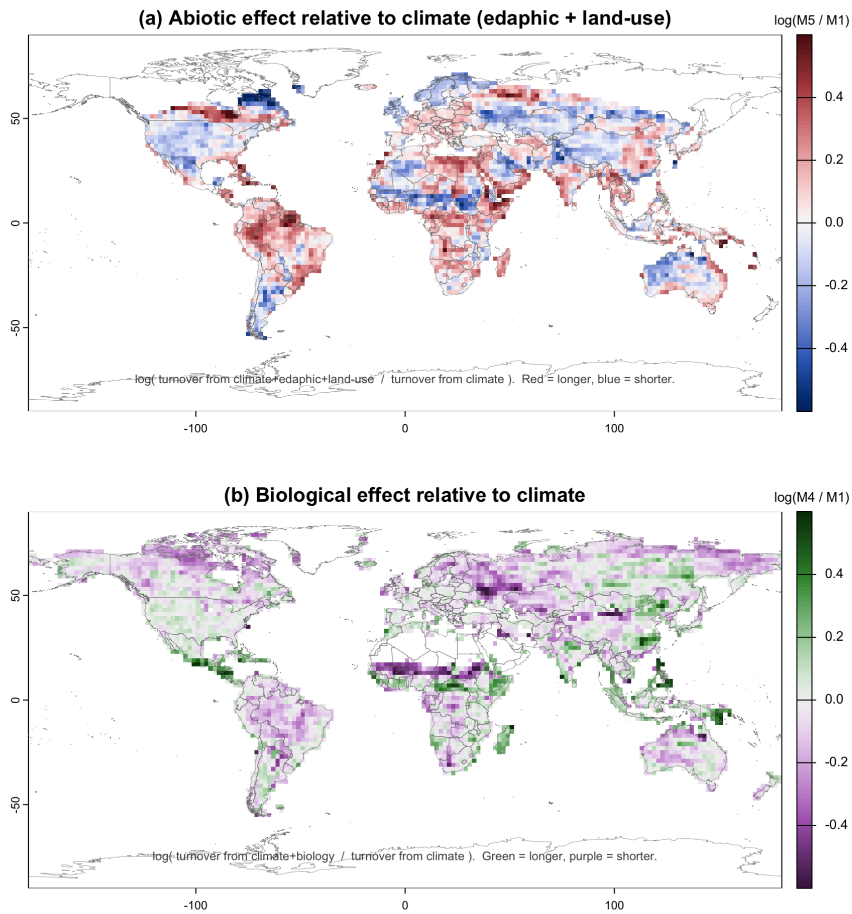

Figure 2: **Zonality map, symmetric (Results).** (a) Abiotic and (b) biological modulation, each relative to climate. Red/green = longer turnover, blue/purple = shorter. The abiotic panel doubles as the near-global headline. The nested formulation and the abiotic-vs-biology scatter are appendix figures. `15b_zonality_map.R`

3.3 Mechanism — “how?” (ALE)

Accumulated Local Effects give the marginal, functional response of turnover to each predictor. The curves are clean and mechanistically sensible: turnover increases with temperature seasonality, snow cover, and elevation; decreases with sand fraction, bulk density, and soil temperature; and is humped in pH. This is the layer that turns the attribution and the map into *mechanism* — it says not just that edaphic matters and where, but *how* each property acts.

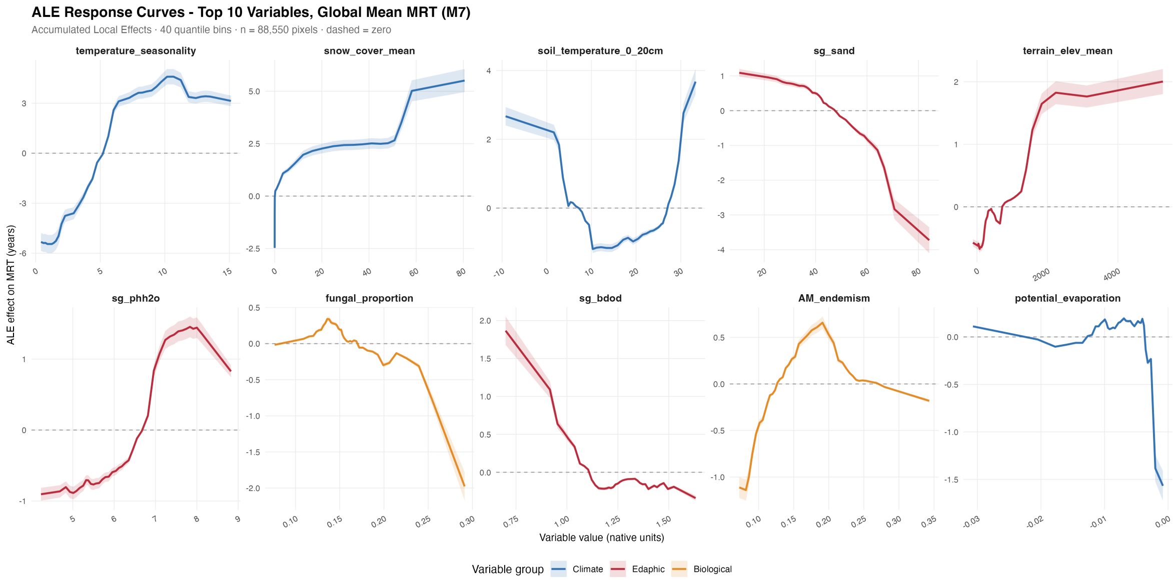

Figure 3: **ALE response curves (Results).** Top-10 predictors, coloured by domain. `18_sensitivity_analysis.R`

3.4 ESM contrast — "so what?"

We compare the spatial *structure* of turnover against six structurally independent CMIP6 models. Absolute turnover times are not comparable across products (depth, pool, and reference-period definitions differ), so we normalise each product to its own global mean and compare structure — a choice that pre-empts the "apples to pears" critique. The result is sharp: observations show a gentle zonality (polar turnover $\sim$2.5$\times$ tropical), whereas the ESM ensemble amplifies it $\sim$5$\times$ (individual models up to $\sim$14$\times$), and the models disagree among themselves across an order of magnitude — one (MPI) even inverts the gradient. The over-zonalization factor (ensemble vs observations) is $\sim$3.6$\times$.

4 The honesty apparatus (robustness; mostly appendix)

These analyses do not carry the argument but they make the modest signal trustworthy. Each is a numbered, repeatable script.

Spatial block cross-validation. All skill is estimated by holding out whole 2° blocks, so reported R^2 is a true out-of-region generalisation number, not interpolation between near-neighbours.

Noise floor. The residual variogram has a nugget/sill of $\sim$0.56 and a range of $\sim$16 km: roughly two thirds of the beyond-climate residual is unstructured noise, and structure lives only at short range. This is why we map at meso scale and never show native-pixel texture.

Provenance control. Only 2.4% of the residual variance lies between the 44 source databases; removing each source's mean barely changes the coarse organisation or covariate skill. The signal is ecological, not a laboratory/compilation artifact. (Conservative, since source is confounded with geography.)

Biological parsimony. The biological signal is multidimensional, but the three Barceló root-colonisation layers are statistically inert (+0.001 unique R^2) while capping coverage. Dropping them lifts the biological footprint from 61% to 96% of sampled rows with no loss of skill; the full nine-layer set is kept as a sensitivity case.

Block-size invariance. Re-running the decomposition with 1° and 5° blocks moves each Shapley

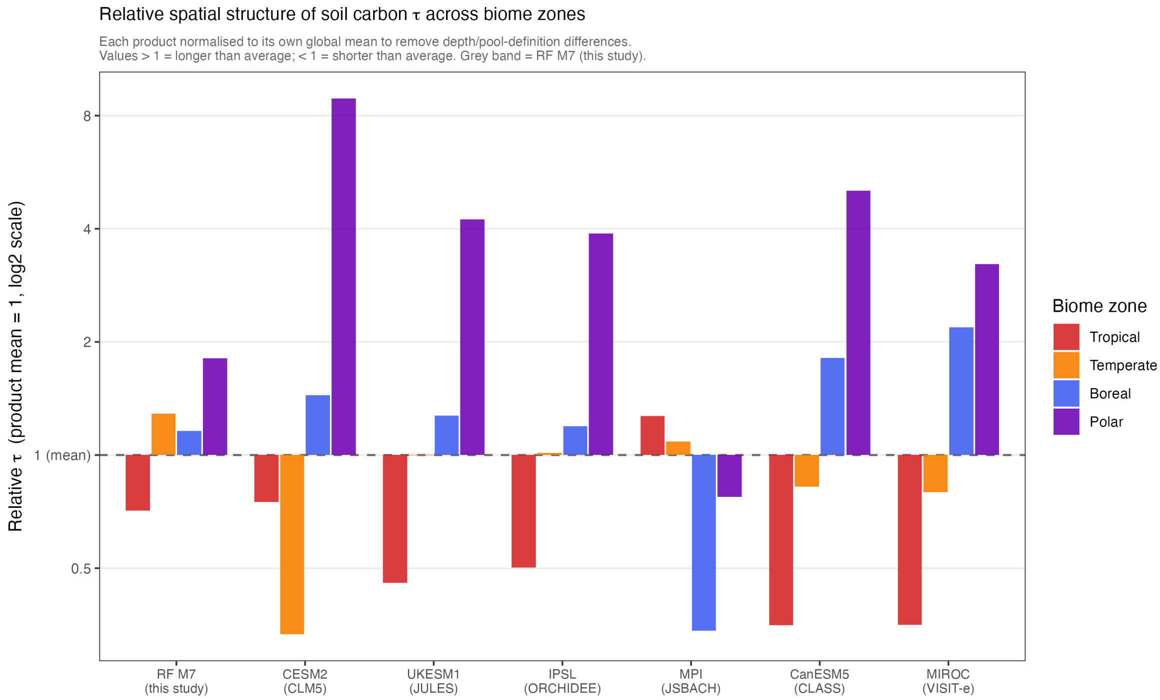

Figure 4: **ESM structural comparison (Results).** Relative turnover by biome zone, each product normalised to its own global mean. `20_CMIP_comparison.R`; amplification ratios `20b_zonal_amplification.R`

share by at most 2.4 percentage points, even though absolute R^2 declines with block size — attribution is robust to the scale choice.

Spatial correlation of the two axes. Abiotic and biological modulation co-vary when each is taken over climate ($r \approx 0.4$–0.5) but are orthogonal once biology is taken conditional on the abiotic model ($r \approx 0$). This is the overlap/"correlated, not redundant" result expressed spatially, and it is its own appendix scatter figure (Köppen-coloured).

5 Key numbers

Quantity	Value
Full-model block-CV R^2	0.38 (climate alone 0.31; beyond-climate ∼7 pp)
Shapley shares (C/E/B/L)	35 / 33 / 22 / 10 %
Single-domain R^2	C 0.31 ≈ E 0.31 > B 0.23 ≫ L 0.11
Overlap (Σ singles − full)	0.57 (correlated, not redundant)
Tier-1 organisation (Moran's I)	0.25 ($p \ll 0.001$)
Noise floor (nugget/sill; range)	0.56; ∼16 km
Between-source variance share	2.4% (ecological, not provenance)
Block-size invariance	shares move ≤2.4 pp over 1–5°
Biological coverage (MAIN set)	96% of rows (vs 61% with Barceló)
Over-zonalization (ESM vs obs)	polar/tropical 2.5× vs ∼9×; factor ∼3.6×

6 Figure plan

7 Two deliberate narrative devices

Own the incomparability. In the ESM section we state up front that absolute turnover is not comparable across products, and therefore compare only normalised spatial structure. Declaring

Figure	Destination	Comment
Zonality map (symmetric)	Results	where; abiotic panel = near-global headline
Variance decomposition	Results	how much + the overlap keystone
ALE response curves	Results	how each driver acts (mechanism)
ESM biome-relative	Results	so what; structural over-zonalization
Tier-1 residual map + Moran	Results/App.	organisation is real, not noise
Scale / noise-floor	Appendix	two-thirds noise; meso-scale justification
Provenance table	Appendix	ecological, not artifact
Block-size sensitivity	Methods	justifies 2° choice
Biological parsimony	Appendix	why drop Barceló; keep 9 as sensitivity
Nested map + abiotic/biology scatter	Appendix	order-dependence; correlated-not-redundant
Relative latitude heatmap	Appendix	continuous-latitude ESM structure
Absolute latitudinal profile	Appendix	the resolved-tension side-arc (below)
Sampling coverage	Methods/Data	where/when sampled
Dropped (obsolete)	—	step-17 M1–M7 maps, all-models comparison, latitude histogram; step-20 absolute heatmap

the limitation first, and controlling for it, converts a potential "apples to pears" objection into a deliberate, rigorous design choice.

Resolved-tension side-arc (Discussion). We briefly raise the absolute-magnitude caveat (with the absolute latitudinal profile relegated to the appendix) and then resolve it via the structural comparison. It is a small arc for robustness, signalling that we considered and dismissed the obvious objection; it is not central to the argument.

8 Status and next steps

Settled. Shapley-primary lead; tool-first framing; symmetric map as the Results centerpiece with the nested version and the scatter in the appendix; the abiotic-vs-biology correlation reported briefly in the Discussion; the ESM section reframed as structural; step-17 demoted, step-18 kept, step-20's absolute profile moved to the appendix side-arc.

Open (small). A minimum-fill coverage threshold for the meso maps; an optional per-pixel Shapley map as the order-free gold standard (needs a step-14 extension); and a final refresh of the (currently 300-tree) block-size sweep at 500 trees.

Consistency item for the post-freeze code polish. The ALE (step 18) and the prediction maps (step 14) were built on the all-nine-layer biology model; for full consistency the final pass should regenerate them with the MAIN 6-layer set. This does not affect the climate/edaphic curves or the abiotic map, so it is cleanup, not a story risk.

Next. The prose pass: draft `manuscript.tex` section by section against this outline and the reconciliation memo, then a code-polish pass once the story is frozen.